

# Andre Arcaina

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## Education

### Toronto Metropolitan University (TMU, formerly Ryerson University)

Sept 2022 – Expected June 2027

Bachelor of Science (Honours), Computer Science (Co-op)

Toronto, ON

**Coursework:** Data Structures, Algorithms, Databases, Computer Networks, Operating Systems, Software Engineering, Object Oriented Programming, Distributed Systems, Big Data Systems

## Work Experience

### Environment and Climate Change Canada (ECCC)

June 2025 – Present

Software Engineer Intern

North York, ON

- Modernized 10+ mission-critical Java services to Java 17 and Jakarta EE 10, improving maintainability and supporting production systems used for national weather, maritime, and Coast Guard operations.
- Utilized Graylog to analyze Java application logs and Grafana to monitor CPU, heap usage, throughput, and data rejections, helping troubleshoot production issues and improve system reliability.
- Improved Jenkins-based continuous integration by developing a Go end-to-end testing framework that validated component data flow across multiple test cases, reducing bug triage time for satellite data pipelines.
- Rebuilt a legacy VB6 data-processing application in Python/PyQt with a PostgreSQL backend, securely processing datasets exceeding 50,000 rows and improving long-term application supportability.
- Refactored 3 legacy C++ services by removing deprecated exception patterns, modernizing error handling, and improving code clarity, reducing technical debt and improving long-term maintainability across production systems.

### Undergraduate Science Society of TMU (USSTM)

Jan 2025 – Apr 2026

Backend Engineer

Toronto, ON

- Architected a campus vault backend for the undergraduate science student body by engineering a contract-first Go RESTful API across 32+ endpoints, leveraging SQLc for type safety and OpenAPI for strict schema enforcement.
- Improved release reliability by implementing 37+ unit and integration tests, achieving over 85% code coverage and preventing regressions in core application workflows.
- Built AWS S3 image storage with presigned URLs to support secure, horizontally scalable before and after item image uploads.
- Developed asynchronous email processing using Redis-backed Asynq workers and AWS SES for reliable notification delivery.

### DataKinetics

May 2024 – Aug 2024

Software Engineer Intern

Ottawa, ON

- Built a Spring Boot REST API and web dashboard with COBOL Copybook to JSON parsing, enabling engine to edit Db2 records and decode EBCDIC/hex data for easier mainframe inspection.
- Created 75 Postman requests with 97 automated JavaScript assertions for a mainframe REST API, validating retrieval, pagination, invalid inputs, error handling, and response data across 3 Db2-backed tables.
- Documented the dashboard architecture and workflows end-to-end in Confluence and presented to 12 R&D stakeholders.

## Projects

### Distributed Stock Trading Platform | Go, Docker, Kubernetes, Helm, Grafana, NATS, PostgreSQL, Redis

GitHub

- Architected an eight microservice Go paper trading platform and load-tested it with 5,000 concurrent simulated users, averaging 1,600+ RPS with zero failures (P95: 42ms).
- Containerized Go microservices with Docker and deployed them to Kubernetes using reusable Helm templates and Minikube.
- Developed Grafana dashboards backed by Prometheus time-series metrics and Loki logs to monitor service health.

### Concurrent Codebase Analysis CLI & SDK | Go, Cobra, Concurrency, GitHub Actions, CI/CD

GitHub

- Engineered a concurrent Go CLI that scanned 46,000+ files in under 0.7 seconds at over 600% CPU utilization, implementing GitHub Actions and GoReleaser CI/CD for tagged releases.

## Skills

**Languages:** Go, Java, C++, Rust, Python, JavaScript, TypeScript, SQL, Bash

**Frameworks & APIs:** Spring Boot, FastAPI, Flask, Node.js, React.js, Next.js, REST, gRPC, GraphQL

**Infrastructure & Tools:** Docker, Kubernetes, Helm, NATS, Jenkins, GitHub Actions, GCP, AWS, Git, Linux, Make

**Databases & Observability:** PostgreSQL, Redis, BadgerDB, Grafana, Prometheus, Loki, Graylog